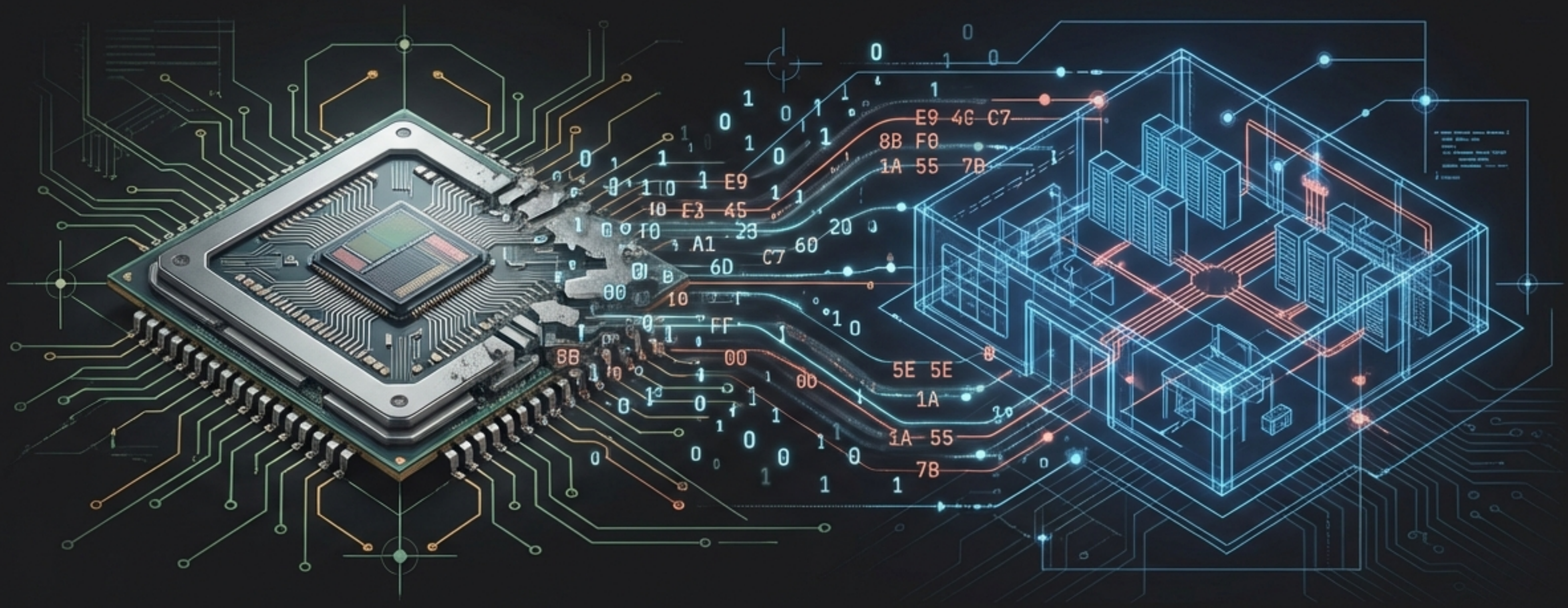


The Art of Reverse Engineering

Deconstructing Software: From Binary to Logic

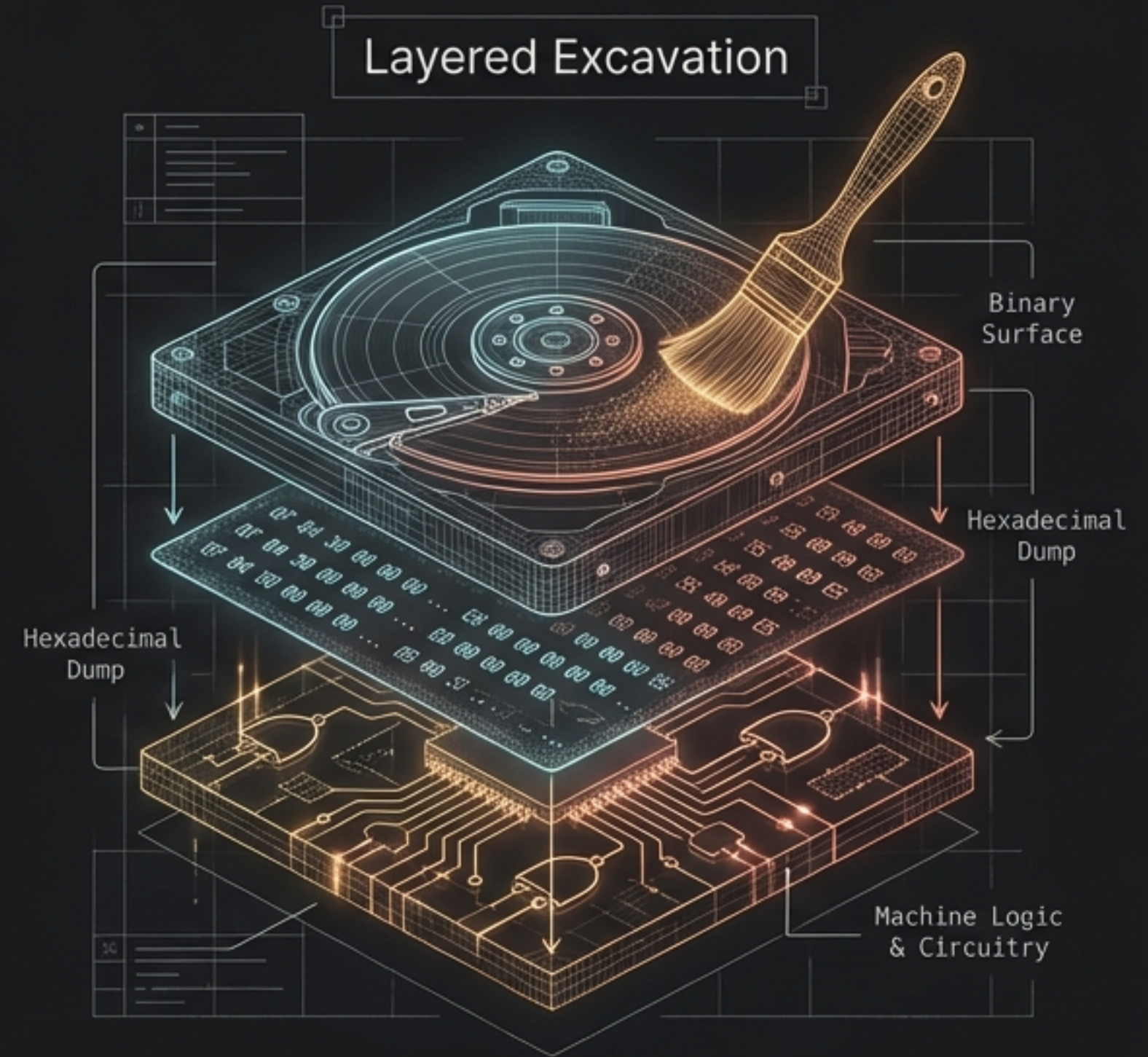


Software Forensics: The Digital Archaeology

Reverse Engineering (RE) is the process of dissecting a compiled binary to reveal its internal logic, algorithms, and design without access to the original source code.

The Black Box Problem

- The Artifact: We have the answer (the binary executable).
- The Mystery: We have lost the question (the source code/intent).

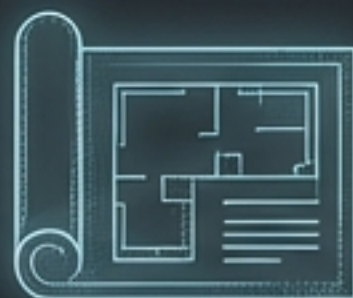


Two Paths to Truth: Static vs. Dynamic

The Analyst must use both the map and the vehicle.

STATIC ANALYSIS

Reading the Blueprint



- **ACTION:** Examining code and metadata on disk without execution.
- **PROS:** Safe (malware is dormant), Complete Coverage (sees all code paths).
- **CONS:** Confused by obfuscation; misses runtime values.
- **ANALOGY:** Studying a map.

DYNAMIC ANALYSIS

Test Driving the Car



- **ACTION:** Executing the program in a controlled debugger.
- **PROS:** Real-time truth (memory values), bypasses static obfuscation.
- **CONS:** Risky (infection chance), Incomplete Coverage (only triggered paths).
- **ANALOGY:** Driving in traffic.

KEY INSIGHT: Static gives you the map; **Dynamic** shows you the traffic.

The Analyst's Arsenal

STATIC TOOLS (The Disassemblers)

1

IDA Pro

The Industry Standard. Interactive, graph-based visualization of control flow.



2

Ghidra

The NSA's open-source alternative. Powerful, scriptable, and free.



DYNAMIC TOOLS (The Debuggers)

3

WinDbg

Kernel-mode and user-mode debugging for Windows deep dives.



4

x64dbg

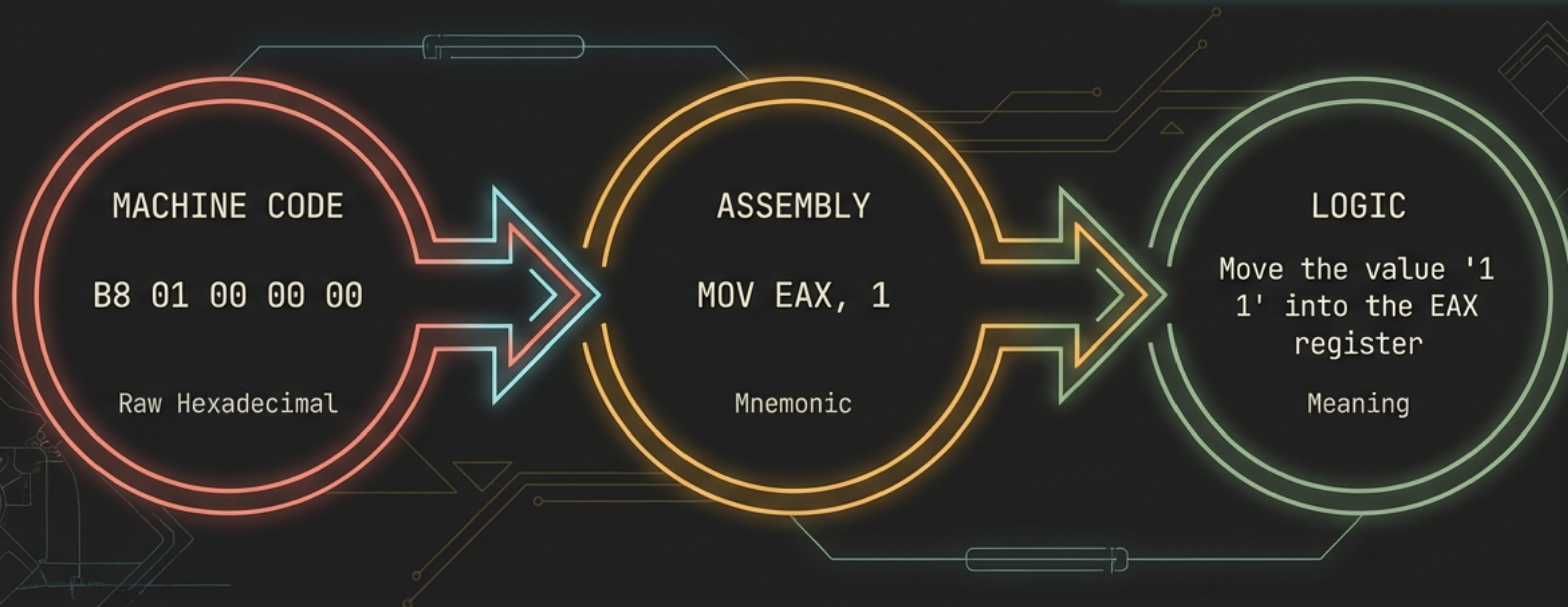
User-friendly, open-source debugger. Preferred for malware analysis.



The Rosetta Stone: Assembly & x86

Binaries are machine code. Assembly is the human translation.

Target: x86 (Windows 32-bit)
Why? Legacy support, malware prevalence, and it is the foundation for x64.



Proficiency in reading Assembly is the single most important skill for a reverse engineer.

The Alignment Challenge

x86 instructions vary in length from 1 to 15 bytes. A disassembler must decode each instruction to find the start of the next.

"One missing byte changes the entire reality of the program."

Data Stream Visualization

A Correct Interpretation



```
MOV EAX, 0x90909090
```

5 Bytes total. Starts at B8.

B Misaligned Interpretation



Catastrophic error. Variable assignment becomes empty noise.

The Origin Story: The Build Process

To reverse it, you must understand how to forward it.



Linking happens on the developer's machine.
Loading happens on the user's machine.

Entropy: The Loss of Context

Compilation is a lossy process. The map is not the territory.

HIGH-LEVEL SOURCE

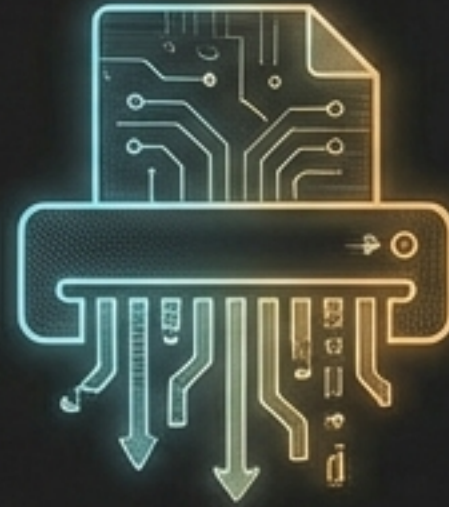
```
struct User {  
    char* name;  
    int id;  
};
```

Data Structures

```
void CheckLogin(User* u) {  
    if (u->id == 0) return;  
}
```

Function Names

Variable Names



DISASSEMBLY

```
sub_401050:  
    MOV    EAX, [ESP+4]  
    CMP    [EAX+4], 0  
    JZ     short loc_401062
```

Generic Address names (sub_401050)

Raw Offsets (+4)

Flattened Logic

What disappears? Names, Types, and Structure.
We must infer them from the bytes.

Algorithmic Intuition: How Tools Guess

Linear Sweep (e.g., WinDbg)



Decodes sequentially. Fast but dumb. Confuses data for code.

Recursive Descent (e.g., IDA Pro)



Follows control flow (Jumps/Calls).
Smart but incomplete. Can miss indirect paths.

Electronic Warfare: Anti-Reverse Engineering

The code fights back.

PACKERS



Compression or encryption layers that hide the payload. The code only "unpacks" itself in memory at runtime.

OBFUSCATION



Spaghetti code.
Inserting junk instructions and logic puzzles to confuse the human analyst.

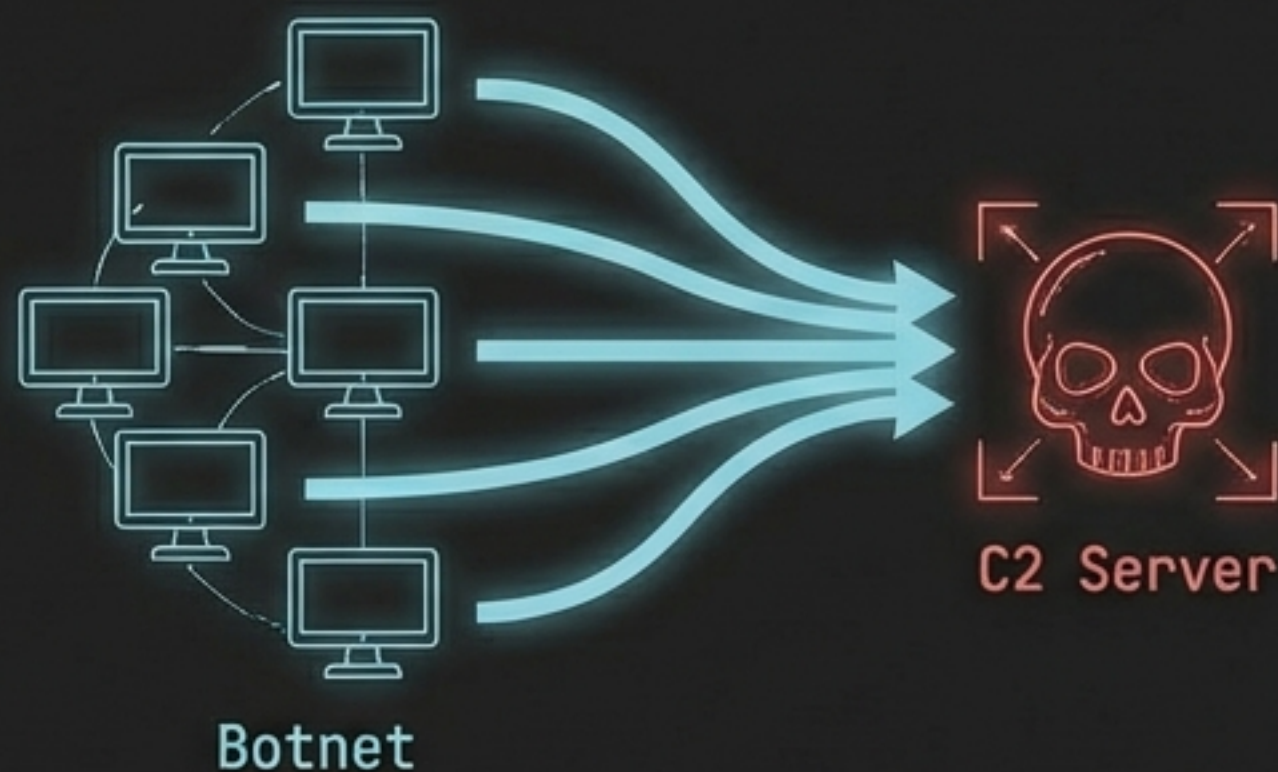
ANTI-DEBUGGING



The code asks:
"IsDebuggerPresent()?".
If yes, it terminates or behaves innocently.

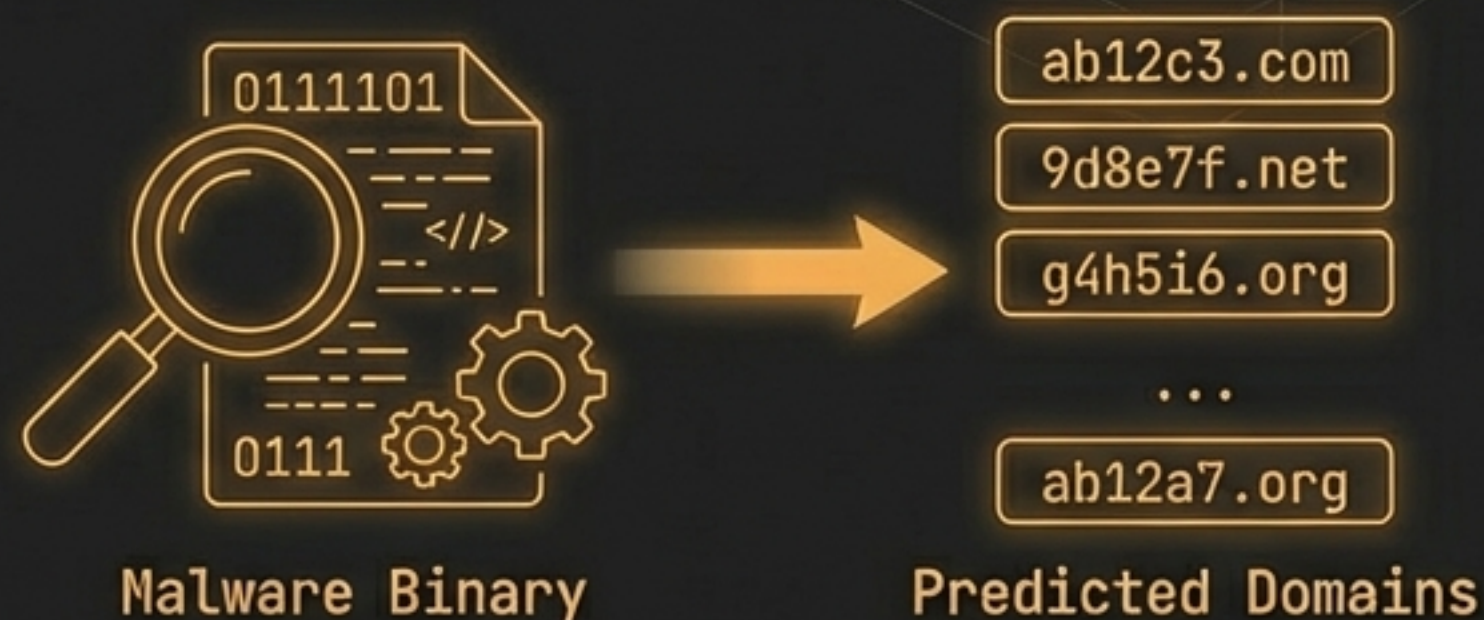
Case Study: Malware & DGAs

The Problem



Domain Generation Algorithms (DGA).
Attackers generate thousands of domains
(e.g., `xy7z1.com`) to hide the real server.

The RE Solution



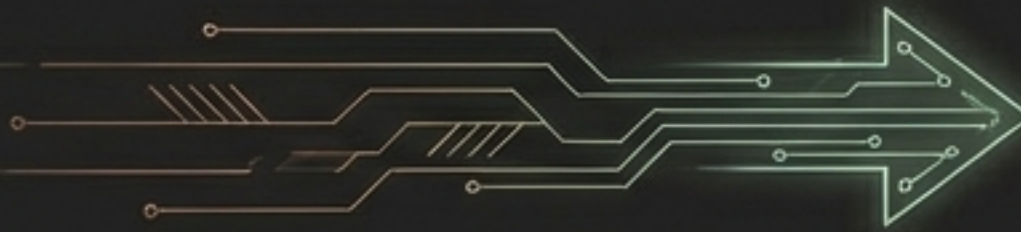
The Analyst reverse engineers the algorithm.

NXDOMAIN Defense.

We predict the domains before they are
registered and block them all.

Beyond Security: Interoperability & Legacy

Interoperability



Opening closed doors. Analyzing raw bytes to decode proprietary formats, allowing open-source tools to communicate with closed systems.

Legacy Maintenance

Critical software where source code is lost. RE allows us to patch bugs directly in the binary to keep systems alive.



The Ghost Ships.

The Road Ahead

Reverse Engineering combines low-level literacy with high-level investigative reasoning.

THE SKILL



Mastering Assembly and compiler behavior.

THE SUPERPOWER



The ability to understand ANY software, with or without documentation.

The binary is not a black box.
It is a puzzle waiting to be solved.

Field Notes: Glossary

STATIC ANALYSIS

Analyzing code without execution
(Safe, Complete).

DYNAMIC ANALYSIS

Analyzing code during execution
(Real-time, Risky).

LINKER

Build-tool that combines object
files into an executable.

LOADER

OS component that maps the
executable into memory.

OPCODE

The machine-language instruction
bytes.

MNEMONIC

The human-readable Assembly
representation (MOV, ADD).

PACKER

Software that compresses/encrypts
an executable to hide logic.

NXDOMAIN

DNS response for a non-existent
domain; used to track DGA malware.